# Lab Assignment 1: Environment Setup, Preprocessing and Exploratory Data Analysis

CSAI2017P: Applied Machine Learning (Lab)

Experiments 1–2 (Units I, IV) • CO1, CO2

**Instructor:** Dr. Tofik Ali **Due:** announced in the lab
**Student Name:** ____________ **SAP ID:** ____________
**Batch:** ________ **Lab quiz:** LQ1, after submission

## Objective

- Stand up a reproducible Python ML environment and prove it works.
- Load a tabular dataset and describe it before modelling anything.
- Diagnose missing values, outliers and scale problems, and treat them.
- **Before the lab:** run `fetch_california_housing()` once at home. It downloads ~1.4 MB and the campus network may block it.

## Datasets used in this assignment

**A1 (California Housing), A2 (Telco Customer Churn).**
Full descriptions, sources and loading snippets for all anchor datasets are in `Anchor-Datasets.pdf`, alongside this brief.

## Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

**Submit:**

- Notebook: `AML_LA01_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

| Section | Level | Choice rule | Marks |
|---|---|---|---|
| A | Easy | Attempt any 2 of 3 (2 marks each) | 4 |
| B | Medium | Attempt any 1 of 2 (4 marks each) | 4 |
| C | Hard | Compulsory | 2 |
| **Total** | | | **10** |

Course outcomes assessed by this assignment: **CO1, CO2**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

## Section A (Easy) — attempt any 2 of 3 $(2 \times 2 = 4)$

**A1. Environment proof** **[2 marks]**
*No dataset.*

(a) Create a fresh environment and install `numpy`, `pandas`, `scikit-learn`, `matplotlib`, `seaborn`, `jupyter`.

(b) Print the version of each of the five libraries and your Python version in one cell.

(c) State in two lines why pinning versions matters when you hand work to someone else.

**A2. First look at the data** **[2 marks]**
*Dataset: A1 (California Housing).*

(a) Load the dataset into a DataFrame and run `head`, `info`, `describe` and `isna().sum()`.

(b) State the number of rows, the number of features, the target, and its units.

(c) Name one feature you suspect is on a very different scale from the others, with evidence.

**A3. Missing values and outliers** **[2 marks]**
*Dataset: A2 (Telco Customer Churn).*

(a) Find the column that looks numeric but is stored as text, and explain why.

(b) Convert it, count how many values become missing, and impute them with a stated strategy.

(c) Draw a boxplot of `tenure` and `MonthlyCharges` and comment on outliers in two lines.

## Section B (Medium) — attempt any 1 of 2 $(1 \times 4 = 4)$

**B1. Scaling changes the answer** **[4 marks]**
*Dataset: A1 (California Housing).*

(a) Split 80/20 with `random_state=0`. Train a `KNeighborsRegressor(n_neighbors=5)` on the raw features and report test MAE.

(b) Repeat inside a `Pipeline` with `StandardScaler`. Report test MAE again.

(c) Explain in 4–6 lines why the difference appears, referring to how KNN measures distance.

**B2. Correlation and leakage** **[4 marks]**

*Dataset: A2 (Telco Customer Churn).*

(a) Encode `Churn` as 0/1 and compute the correlation of every numeric feature with it.

(b) Rank the features and report the three strongest.

(c) `TotalCharges` is roughly `tenure`×`MonthlyCharges`. Explain in 4–6 lines what problem that redundancy causes, and what you would do about it.

## Section C (Hard) — compulsory (2)

**C1. Reproducibility check** **[2 marks]**

*Either dataset.*

(a) Restart the kernel and run your notebook top to bottom.

(b) State whether every number in your report reproduced exactly. If not, identify which and why.

*End of Lab Assignment 1*

*Sources:* Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.